Computer Vision I

Bjoern Andres, David Stein

Machine Learning for Computer Vision
TU Dresden



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Definition 1. For any standard system defined by $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$,

$$A\,x = b \tag{1}$$
$$x \geq 0 \tag{2}$$

and any $c \in \mathbb{R}^n$, we call

$$\min \{\langle c, x \rangle \mid A\,x = b \wedge x \geq 0 \wedge x \in \mathbb{R}^n\} \tag{3}$$

the **linear optimization problem** or **linear program (LP)** in **standard form** and

$$X_{Ab} := \{x \in \mathbb{R}^n \mid A\,x = b \wedge x \geq 0\} \tag{4}$$

its **feasible set**. The elements of X_{Ab} are called the **feasible solutions** to (1)–(2).

Remark 1. Assume $\text{rank}(A) = m$. If $n \leq m$ then $n = m$. In this case, the solution $x' := A^{-1}b$ to (1) is unique. If $x' \geq 0$, $X_{Ab} = \{x'\}$. Otherwise, $X_{Ab} = \varnothing$. More interesting is the case where $n > m$.

Definition 2. For any $m, n \in \mathbb{N}_0$ and any $A \in \mathbb{R}^{m \times n}$ with $\text{rank}(A) = m$, a **basis** of A is any $K \subseteq n$ with $|K| = m$ such that $A|_{\cdot K}$ is non-singular.

Definition 3. For any $m, n \in \mathbb{N}_0$, any $A \in \mathbb{R}^{m \times n}$ with $\text{rank}(A) = m$, any $b \in \mathbb{R}^m$, any $x \in X_{Ab}$ and any basis K of A, x is called a **basic feasible solution (BFS)** with basis K to the standard system defined by A and b iff

$$x|_K = A|_{\cdot K}^{-1} b \tag{5}$$

$$x|_{n \setminus K} = 0 \tag{6}$$

Example:
Consider the standard system $Ax = b$, $x \geq 0$, where

$$A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}, \qquad b = \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

Every pair of columns of A is linearly independent. Thus, the bases are $\{1, 2\}$, $\{1, 3\}$ and $\{2, 3\}$.

For each basis K, set the nonbasic variables to zero and solve $Ax = b$:

K	Nonbasic variable	x	BFS?
$\{1, 2\}$	$x_3 = 0$	$(1, 2, 0)^\top$	Yes
$\{1, 3\}$	$x_2 = 0$	$(-1, 0, 2)^\top$	No: $x_1 < 0$
$\{2, 3\}$	$x_1 = 0$	$(0, 1, 1)^\top$	Yes

The vector $x = (\frac{1}{2}, \frac{3}{2}, \frac{1}{2})^\top$ is feasible but **not basic**: it has three positive components, whereas every basis contains only two indices.

Theorem 1. Let $m, n \in \mathbb{N}_0$, $A \in \mathbb{R}^{m \times n}$ with rank$(A) = m$ and $b \in \mathbb{R}^m$. If $X_{Ab} \neq \varnothing$, there exists a BFS $x \in X_{Ab}$.

Proof. For any $y \in \mathbb{R}^n$, let $K_y := \{k \in n \mid y_k \neq 0\}$. Consider any $x \in X_{Ab}$.

a) If the columns of $A|_{.K_x}$ are l.i., $\text{rank}(A) = m$ implies $|K_x| \leq m$.

- If $|K_x| = m$, x is a BFS with the basis K_x.
- If $|K_x| < m$, choose $K \subseteq n$ such that $K_x \subseteq K$ and $|K| = m$ such that the columns of $A|_{.K}$ are l.i.. This is possible because $\text{rank}(A) = m$. Now, x is a BFS with the basis K.

b) If the columns of $A|_{.K_x}$ are l.d., there exists $d \in \mathbb{R}^n$ such that

$$A\,d = 0 \tag{7}$$

$$\varnothing \neq K_d \subseteq K_x \ . \tag{8}$$

For any $\lambda \in \mathbb{R}$, let $x' := x + \lambda\,d$. Now:

$$A\,x' = A\,x + \lambda\,A\,d \overset{(7)}{=} A\,x = b \ . \tag{9}$$

Choose $k' \in K_d$ so as to minimize $\frac{x_{k'}}{|d_{k'}|}$. Let $\lambda' := -\frac{x_{k'}}{d_{k'}}$. Now:

$$\forall k \in n \setminus K_x: \quad x'_k = x_k + \lambda' d_k \overset{(8)}{=} 0 \tag{10}$$

$$x'_{k'} = x_{k'} - \frac{x_{k'}}{d_{k'}} d_{k'} = 0 \tag{11}$$

$$\forall k \in K_x \setminus \{k'\}: \quad x'_k = x_k - \frac{x_{k'}}{d_{k'}} d_k = \begin{cases} x_k & \text{if } d_k = 0 \\ \left(\frac{x_k}{d_k} - \frac{x_{k'}}{d_{k'}}\right) d_k & \text{if } d_k \neq 0 \ . \end{cases} \tag{12}$$

Proof (contd.). We analyze (12): If $d_k > 0$ then

$$\left(\frac{x_k}{d_k} - \frac{x_{k'}}{d_{k'}}\right) d_k \;=\; \left(\frac{x_k}{|d_k|} - \frac{x_{k'}}{d_{k'}}\right) |d_k| \;\geq\; \left(\frac{x_k}{|d_k|} - \frac{x_{k'}}{|d_{k'}|}\right) |d_k| \;\geq\; 0 \;. \tag{13}$$

If $d_k < 0$ then

$$\begin{aligned}\left(\frac{x_k}{d_k} - \frac{x_{k'}}{d_{k'}}\right) d_k \;&=\; \left(-\frac{x_k}{d_k} + \frac{x_{k'}}{d_{k'}}\right) |d_k| \;=\; \left(\frac{x_k}{|d_k|} + \frac{x_{k'}}{d_{k'}}\right) |d_k| \\ &\geq\; \left(\frac{x_k}{|d_k|} - \frac{x_{k'}}{|d_{k'}|}\right) |d_k| \;\geq\; 0 \;.\end{aligned} \tag{14}$$

Thus, $x' \geq 0$. Together with (9) follows $x' \in X_{Ab}$. Moreover, $K_{x'} \subseteq K_x \setminus \{k'\}$.

If the columns of $A|_{\cdot K_{x'}}$ are l.i., a BFS exists by Part (a).

Otherwise, Part (b) can be iterated until the columns of $A|_{\cdot K_{x'}}$ are l.i..

Thus, a BFS always exists. □

In the remainder of this exercise, we will introduce the **simplex algorithm**:

- The simplex algorithm finds an optimal solution LPs and works well in practice but has exponential worst-case time complexity.
- Other (more complicated) polynomial time algorithms for solving LPs exist.
- The simplex algorithm is based roughly on the idea of moving from one basis to another "adjacent" basis with a lower cost until no further cost reduction is possible.
- The question is how to move from one BFS to an "adjacent" basis and how to find one with a lower cost.

Definition 4. Let $m, n \in \mathbb{N}_0$, $A \in \mathbb{R}^{m \times n}$ with $\text{rank}(A) = m$ and $b \in \mathbb{R}^m$. For

- any BFS x with basis K to the standard system defined by A and b
- any $k^* \in n \setminus K$ and the unique $y^{k^*} \in \mathbb{R}^n$ such that

$$\sum_{k \in K} y_k^{k^*} A_{\cdot k} = A_{\cdot k^*} \qquad \text{i.e. } y^{k^*}|_K = A|_{\cdot K}^{-1} A_{\cdot k^*} \tag{15}$$

$$\forall k \in n \setminus K: \qquad y_k^{k^*} = 0 \tag{16}$$

- any $k^\dagger \in K_{k^*} := \{k \in K \mid y_k^{k^*} > 0\}$ and the unique $\lambda' \in \mathbb{R}_0^+$ such that

$$\min \left\{ \frac{x_k}{y_k^{k^*}} \;\middle|\; k \in K_{k^*} \right\} = \frac{x_{k^\dagger}}{y_{k^\dagger}^{k^*}} = \lambda' \ , \tag{17}$$

the $x' \in \mathbb{R}^n$ such that

$$x'_{k^*} := \lambda' \tag{18}$$

$$\forall k \in n \setminus \{k^*\}: \quad x'_k := x_k - \lambda' y_k^{k^*} \tag{19}$$

is called the result of **pivoting** x in K such that k^* **enters** and $k^\dagger$ **leaves** the basis.

Theorem 2. Let

- $m, n \in \mathbb{N}_0$, $A \in \mathbb{R}^{m \times n}$ with rank$(A) = m$ and $b \in \mathbb{R}^m$
- x a BFS with basis K to the standard system defined by A and b
- x' be the result of pivoting x in K such that k^* enters and $k^\dagger$ leaves the basis.

Then, x' is a BFS with basis $(K \setminus \{k^\dagger\}) \cup \{k^*\}$ to the standard system defined by A and b.

Proof. Since x is a BFS with basis K:

$$\sum_{k \in K} x_k \, A_{\cdot k} = b \ . \tag{20}$$

For any $\lambda \in \mathbb{R}$, (20)–λ(15) yields

$$\lambda A_{\cdot k^*} + \sum_{k \in K} (x_k - \lambda y_k^{k^*}) \, A_{\cdot k} = b \ . \tag{21}$$

For $\lambda = \lambda'$ specifically follows $\mathbf{A}\,\mathbf{x}' = \mathbf{b}$.

From $x_{k\dagger} \geq 0$ and $y_{k\dagger}^{k^*} > 0$ follows $\lambda' \geq 0$. Thus $x'_{k^*} = \lambda' \geq 0$. For any $k \in n \setminus \{k^*\}$, distinguish two cases: If $y_k^{k^*} \leq 0$ then (19), $x_k \geq 0$ and $\lambda' \geq 0$ imply $x'_k \geq 0$. If $y_k^{k^*} > 0$ then

$$x'_k = x_k - \lambda' y_k^{k^*} = x_k - \frac{x_{k\dagger}}{y_{k\dagger}^{k^*}} y_k^{k^*} = \left(\frac{x_k}{y_k^{k^*}} - \frac{x_{k\dagger}}{y_{k\dagger}^{k^*}} \right) y_k^{k^*} \overset{(17)}{\geq} 0 \ . \tag{22}$$

Thus, $\mathbf{x}' \geq \mathbf{0}$.

It remains to be shown that $(K \setminus \{k^\dagger\}) \cup \{k^*\}$ is a basis. ...

Proof (contd.). Let $c \in \mathbb{R}^{(K \setminus \{k^\dagger\}) \cup \{k^*\}}$ such that

$$0 = \sum_{k \in (K \setminus \{k^\dagger\}) \cup \{k^*\}} c_k \, A_{\cdot k} \quad . \tag{23}$$

Now,

$$0 = c_{k^*} A_{\cdot k^*} + \sum_{k \in K \setminus \{k^\dagger\}} c_k A_{\cdot k} \tag{24}$$

$$\overset{(15)}{=} c_{k^*} \sum_{k \in K} y_k^{k^*} A_{\cdot k} + \sum_{k \in K \setminus \{k^\dagger\}} c_k A_{\cdot k} \tag{25}$$

$$= c_{k^*} y_{k^\dagger}^{k^*} A_{\cdot k^\dagger} + \sum_{k \in K \setminus \{k^\dagger\}} (c_{k^*} y_k^{k^*} + c_k) A_{\cdot k} \quad . \tag{26}$$

The rhs. is a l.c. of the columns of A indexed by the basis K. Thus,

$$c_{k^*} y_{k^\dagger}^{k^*} = 0 \tag{27}$$

$$\forall k \in K \setminus \{k^\dagger\}: \quad c_{k^*} y_k^{k^*} + c_k = 0 \quad . \tag{28}$$

From (27) and $y_{k^\dagger}^{k^*} > 0$ follows $c_{k^*} = 0$. Together with (28) follows $c = 0$. Thus, $(K \setminus \{k^\dagger\}) \cup \{k^*\}$ is a basis. □

Example:
Consider

$$A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}, \qquad b = \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

Start with basis $K = \{1, 2\}$ and BFS $x = (1, 2, 0)^\top$.

Let $k^* = 3$ enter the basis. Since $A_{\cdot 3} = A_{\cdot 1} + A_{\cdot 2}$,

$$y^3 = (1, 1, 0)^\top, \qquad \lambda' = \min\left\{ \frac{x_1}{y_1^3}, \frac{x_2}{y_2^3} \right\} = \min\{1, 2\} = 1.$$

The minimum is attained at $k^\dagger = 1$, so variable 1 leaves the basis.

The pivot gives

$$x' = \begin{pmatrix} x_1 - \lambda' y_1^3 \\ x_2 - \lambda' y_2^3 \\ \lambda' \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \qquad K' = (K \setminus \{1\}) \cup \{3\} = \{2, 3\}.$$

Thus, x' is a BFS with the new basis K'.

Lemma 1. Consider

- any $m, n \in \mathbb{N}_0$, $b \in \mathbb{R}^m$, $c \in \mathbb{R}^n$ and $A \in \mathbb{R}^{m \times n}$ such that $\text{rank}(A) = m$
- any BFS x with basis K to the standard system defined by A and b
- any $k^* \in n \setminus K$ and the $y^{k^*} \in \mathbb{R}^n$ with $y^{k^*}|_K = A|_{\cdot K}^{-1} A_{\cdot k^*}$ and $y^{k^*}|_{n \setminus K} = 0$
- x' the result of pivoting x in K such that k^* enters the basis.

Then:

$$\langle c, x' \rangle - \langle c, x \rangle = \left(c_{k^*} - \sum_{k \in K} c_k y_k^{k^*} \right) x'_{k^*} \tag{29}$$

Definition 5. Under the assumptions of Lemma 1, the **relative cost** of k^* in the basis K is the number

$$c_{k^*} - \sum_{k \in K} c_k y_k^{k^*} \quad . \tag{30}$$

Proof. By definition of pivoting, $k^* \notin K$. Thus $x_{k^*} = 0$. Consequently:

$$\begin{aligned}
\langle c, x' \rangle - \langle c, x \rangle &= \langle c, x' - x \rangle && (31)\\
&= \sum_{k \in K \cup \{k^*\}} c_k (x'_k - x_k) && (32)\\
&= c_{k^*}(x'_{k^*} - 0) + \sum_{k \in K} c_k (x'_k - x_k) && (33)\\
&\overset{(19)}{=} c_{k^*} x'_{k^*} + \sum_{k \in K} c_k (-x'_{k^*} y_k^{k^*}) && (34)\\
&= \left(c_{k^*} - \sum_{k \in K} c_k y_k^{k^*} \right) x'_{k^*} \ . && (35)
\end{aligned}$$

□

Theorem 3. Let

- $m, n \in \mathbb{N}_0$, $b \in \mathbb{R}^m$, $c \in \mathbb{R}^n$ and $A \in \mathbb{R}^{m \times n}$ such that rank$(A) = m$
- x a BFS with basis K to the standard system defined by A and b
- for any $k \in n$, c'_k the relative cost of k in the basis K, and $c'_k \geq 0$.

Then, x is an optimal solution to the LP in standard form defined by A, b and c.

Proof. Let x' be any feasible solution to the standard system defined by A and b, not necessarily basic. Now:

$$
\begin{aligned}
\langle c, x' \rangle &= \sum_{k \in n} c_k x'_k \\
&\geq \sum_{k \in n} \left(\sum_{k' \in K} c_{k'} y^k_{k'} \right) x'_k \\
&= \sum_{k' \in K} c_{k'} \sum_{k \in n} y^k_{k'} x'_k = \sum_{k' \in K} c_{k'} \sum_{k \in n} (A|^{-1}_{\cdot K} A_{\cdot k})_{k'} x'_k \\
&= \sum_{k' \in K} c_{k'} \sum_{k \in n} \left(\sum_{j \in m} (A|^{-1}_{\cdot K})_{k'j} A_{jk} \right) x'_k \\
&= \sum_{k' \in K} c_{k'} \sum_{j \in m} (A|^{-1}_{\cdot K})_{k'j} \sum_{k \in n} A_{jk} x'_k = \sum_{k' \in K} c_{k'} \sum_{j \in m} (A|^{-1}_{\cdot K})_{k'j} (Ax')_j \\
&= \sum_{k' \in K} c_{k'} \sum_{j \in m} (A|^{-1}_{\cdot K})_{k'j} b_j = \sum_{k' \in K} c_{k'} x_{k'} \\
&= \langle c, x \rangle
\end{aligned}
$$

□

Input: a BFS x^0 with basis K

$t := 0$
while true
    for any $k \in n$, let c'_k the relative cost of k in the basis K
    if $c' \geq 0$
        return x^t (solution)
    else
        choose k^* such that $c'_{k^*} < 0$
        if $K_{k^*} = \varnothing$
            return unbounded
        else
            choose $k^\dagger$ and define λ' according to (17)
            let x^{t+1} the pivoting of x^t s.t. k^* enters and $k^\dagger$ leaves K
            $t := t + 1$

Simplex Algorithm

Input: a BFS x^0 with basis K

$t := 0$
while true
    for any $k \in n$, let c'_k the relative cost of k in the basis K
    if $c' \geq 0$
        return x^t (solution)
    else
        choose k^* such that $c'_{k^*} < 0$
        if $K_{k^*} = \varnothing$
            return unbounded
        else
            choose $k^\dagger$ and define λ' according to (17)
            let x^{t+1} the pivoting of x^t s.t. k^* enters and $k^\dagger$ leaves K
            $t := t + 1$

Simplex Algorithm

Pivoting rules:

– Choose k^* so as to minimize the relative cost

$$c_{k^*} - \sum_{k \in K} c_k y_k^{k^*} \quad . \tag{36}$$

Not guaranteeing locally optimal reduction in cost; not preventing cycling.

– Choose k^* so as to minimize

$$\left(c_{k^*} - \sum_{k \in K} c_k y_k^{k^*} \right) x'_{k^*} \quad . \tag{37}$$

Guaranteeing locally optimal reduction in cost; not preventing cycling.

– Bland's Rule: Define a total order $<$ on n. Among those k^* with a negative reduced cost, choose the one that is smallest wrt. $<$. If there is a tie in (17), choose $k^\dagger$ to be the smallest wrt. $<$.

Not guaranteeing locally optimal reduction in cost; preventing cycling.

We have considered linear programs (LPs). An **integer linear program (ILP)** additionally requires its variables to take integer values:

$$z_{\mathrm{ILP}} := \min\bigl\{\langle c, x\rangle \mid Ax = b,\ x \geq 0,\ x \in \mathbb{Z}^n\bigr\}.$$

Solving ILPs is **NP-hard** in general.

Dropping the integrality constraints gives the **LP relaxation**:

$$z_{\mathrm{LP}} := \min\bigl\{\langle c, x\rangle \mid Ax = b,\ x \geq 0,\ x \in \mathbb{R}^n\bigr\}.$$

- The LP relaxation can be solved in polynomial time (in practice often using the simplex algorithm).
- Every feasible solution of the ILP is feasible for the LP relaxation. Therefore,

$$\boxed{z_{\mathrm{LP}} \leq z_{\mathrm{ILP}}}.$$

  The LP relaxation provides a **lower bound** on the optimal ILP objective value.